CIE

COPYRIGHT INFORMATION EXTRACTION FORM

TITLE: **CIE-Vault**

SCHOOL: Chitkara University, Rajpura, Punjab

DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS

Name:

Roll No/ Employee Code:

All MEMBERS DETAILS INCLUDING SUPERVISOR

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ABSTRACT

The process of moving from research to intellectual property filing can be a challenge for students, especially regarding to the unorganized handling of filing receipts and the insecure sharing of government portal credentials with supervisors. To solve these problems, CIE-Vault (Copyright Information Extraction) is suggested as a niche MERN-stack solution to organize the copyright process in academic settings. The main goal of the system is to offer a secure and centralized repository for data entry automation, along with a complex and institution-conscious access control system. In essence, CIE-Vault utilizes Node.js and the pdf-parse library to enable Automated Extraction. Upon a student submitting a copyright filing receipt, the system automatically parses the document to extract essential information, such as "Filing Number" and "Title," thus eliminating the possibility of human error. In addition to mere storage, the system incorporates a Hybrid Access Control system that differentiates between internal and external mentorship. The "Smart" system is regulated by College-Based Routing; upon registration, users are assigned to particular colleges, thus establishing a virtual boundary that determines the visibility of the data. The system provides a dual-role setting for Students and Mentors, which is ensured through JWT (JSON Web Tokens) and Bcrypt encryption. The student interface enables the seamless upload of abstracts and the safe storage of portal passwords in a secure "Vault." The novelty of CIE-Vault is in its logic of permissions. Internal Mentors, who are from the same college as the student, are assigned "Full Access," allowing them to access confidential filing documents and portal credentials. On the other hand, External Mentors from other colleges are peer reviewers with a "Global Research" tab. These external reviewers are restricted to accessing only the abstract of the project and offering "Expert Suggestions," while ensuring that the intellectual property is reviewed by the broader academic community without accessing the student's private login information or confidential filing receipts. With the incorporation of MongoDB for dynamic data structuring and Tailwind CSS for a responsive and contemporary design, CIE-Vault fills the existing gap between administrative compliance and collaborative research. It eliminates disorganized communication in favour of a systematic workflow that promotes institutional security alongside inter-collegiate feedback. Ultimately, CIE-Vault acts as a powerful tool for academic institutions seeking to professionalize their intellectual property processes, ensuring that every copyright filing is properly tracked, secured, and analyzed by the relevant stakeholders.

Problem Background:

Academic copyright submissions handled manually are usually associated with fragmented information and substantial security risks. Presently, students are known to share government portal login credentials with their mentors through insecure means such as email or instant messaging. This absence of a systematic repository results in lost documents, manual errors in data entry, and a general absence of transparency in the submission process. Moreover, there is no systematic way for institutions to enable peer review from outside sources without compromising the private details of the students.

Objective of the Project:

The main goal of CIE-Vault is to ensure a secure and automated system for managing copyright filings. The system seeks to make data extraction easier through PDF parsing, ensure sensitive credentials are protected by hybrid access control, and make collaboration between students and academic mentors easier.

Proposed Solution / System Overview:

CIE-Vault is a niche MERN-stack solution developed to automate the entire lifecycle of academic copyright management by using a single [Automated Extraction] service. The key components of the system are Dual-Role Authentication, College-Based Routing, and a Hybrid Access Control engine, which interact with each other to facilitate copyright documentation and ownership verification. With the use of Node.js and pdf-parse, the system allows instant extraction of filing numbers from the uploaded receipts, and the Institutional Logic module ensures that the portal credentials are visible only to the internal, authenticated mentors.

```
cie-vault/
├── client/ (React + Tailwind)
│   ├── src/
│   │   ├── components/  # Reusable UI (Navbar, Sidebar)
│   │   ├── pages/      # Login, StudentDashboard, MentorDashboard
│   │   └── context/    # Auth & Permission state
├── server/ (Node + Express)
│   ├── controllers/    # Extraction, Auth, and File Logic
│   ├── middleware/     # JWT and Permission Checkers
│   ├── models/         # MongoDB Schemas
│   └── utils/          # PDF Parser Utility
└── uploads/            # Secure Storage for PDF Receipts
```

Key Features or Functionality:

CIE-Vault integrates Automated Extraction, featuring **pdf-parse** to immediately extract filing numbers and project names from uploaded files. Hybrid Access Control is a central feature, employing institutional logic to separate internal and external reviewers. The system also features a Secure Portal Vault for encrypted credential storage and Dual-Role Dashboards, enabling students to choose college-specific advisors while facilitating global research feedback via a limited, abstract-only viewing mode for external academic reviewers.

Technology / Method Used:

CIE-Vault is developed using the MERN stack, leveraging React.js with Tailwind CSS for a responsive frontend and Node.js with Express.js for the backend. MongoDB is used for data management, while JWT and Bcrypt provide authentication security. pdf-parse is incorporated for automatic document metadata extraction.

Expected Outcome & Benefits:

The application of CIE-Vault leads to a more streamlined and secure environment for the administration of institutional intellectual property. The benefits include the elimination of the need for manual data entry with automated extraction, as well as the elimination of risks associated with the sharing of credentials. Additionally, the application of CIE-Vault leads to improved data integrity through PDF parsing, privacy through hybrid access control, and a repository that promotes accountability as well as collaboration.

Original Contribution Statement:

CIE-Vault proposes a novel Hybrid Access Control model and an Automated Extraction process, combining for the first time the management of secure credentials with research collaboration between institutions.